NOT MEASUREMENT SENSITIVE

MIL-A-12545D(MR)
16 June 1989
SUPERSEDING
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8 December 1972

# MILITARY SPECIFICATION

# ALUMINUM ALLOY IMPACTS

This specification is approved for use by the Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1. Scope. This specification covers five aluminum alloys in the form of impacts.

1.2 Classification.

1.2.1 Alloys and tempers. Impacts furnished under this specification shall be of the following alloys and tempers, as specified (see 6.1):

*1100-F - As fabricated.*

2014-F - As fabricated.
-O - Annealed.
-T4 - Solution heat treated.
-T6 - Solution heat treated and artifically aged.

6061-F - As fabricated.
-O - Annealed.
-T4 - Solution heat treated.
-T6 - Solution heat treated and artifically aged.
-T84 - Solution heat treated, cold worked, and artifically aged.

6070-F - As fabricated.
-T6 - Solution heat treated and artifically aged.

7075-F - As fabricated.
-O - Annealed.
-T6 - Solution heat treated and artifically aged.
*-T73 - Solution heat treated and artifically aged to develop* mechanical properties and resistance to stress-corrosion cracking.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Director, U.S. Army Laboratory Command, Materials Technology Laboratory, ATTN: SLCMT-MEE, Watertown, MA 02172-0001 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 95GP
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1.2.2 Alloy and temper designations are in accordance with ANSI H35.1. The equivalent UNS alloy designations are those of Table I preceded by A9, for example, A91100 for aluminum 1100 in accordance with practice E527.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.1).

STANDARDS

MILITARY

MIL-STD-649 - Aluminum and Magnesium Products, Preparation for Shipment and Storage

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.1).

ASTM (AMERICAN SOCIETY FOR TESTING AND MATERIALS)

ASTM B557 - Tension Testing Wrought and Cast Aluminum and Magnesium Alloy Products
ASTM B660 - Practices for Packaging/Packing of Aluminum and Magnesium Products
ASTM E10 - Brinell Hardness of Metallic Materials
ASTM E34 - Standard Test Methods for Chemical Analysis of Aluminum and Aluminum Base Alloys
ASTM E527 - Practice for Numbering Metals and Alloys (UNS)

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103-1137.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI H35.1 - Alloy and Temper Designations Systems for Aluminum

(Application for copies should be addressed to ANSI, 1430 Broadway, New York, NY 10018)



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(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material. Impacts shall be produced from disks, blanks, or slugs of suitable shape and dimensions.

3.2 Chemical composition.

3.2.1 The chemical composition of the impacts shall be as specified in table I.

3.2.2. Analysis shall be made routinely only for the elements specifically stated in table I. If the presence of other elements is suspected or indicated in the course of the analysis, further analysis shall be made to determine conformance with the limits specified for other elements.

3.3 Mechanical property requirements.

3.3.1 Except as provided in 3.3.4, mechanical properties shall be determined by tension tests of the sidewall of the impact.

3.3.2 Tensile properties of impacts. Tension test specimens representative of the impacts shall meet the requirements shown in table II for the temper specified.

3.3.3 Tensile properties after heat treatment. In addition to the requirements of 3.3.2, when impacts other than 1100 alloy are ordered in the annealed or as fabricated temper, representative samples shall be solution heat treated and aged and shall meet the requirements shown in table II for the solution heat treated and artificially aged temper of the alloy ordered.

3.3.4 Hardness. When it is not practicable to obtain standard or sub-sized tension test specimens from the sidewall of the impact, the hardness requirements shown in table II, determined from readings taken on the base, shall be substituted for the tensile property requirements specified in 3.3.2 and 3.3.3.

3.4 Dimensional tolerances. The impacts shall conform to the shape and dimensions specified in the contract or purchase order (see 6.1) within such variations as may be shown on the drawings.

3.5 Workmanship. The impacts shall be of uniform quality and condition and shall be free from blisters, scratches, and other defects that would affect their serviceability, within the limits that are consistent with best commercial practices. Discoloration due to heat treatment shall not be cause for rejection.



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TABLE I. Chemical composition, percent by weight.

| Alloy | Silicon | Iron max. | Copper | Manganese | Magnesium | Chromium | Zinc | Titanium, max. | Other Elements Each, max. | Other Elements Total, max. | Aluminum |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1100 | 1/ | 1/ | 0.05 0.20 | 0.05 max. | - | - | 0.10 max. | - | 0.05 | 0.15 | 99.00 min. |
| 2014 | 0.50-1.20 | 0.70 | 3.90-5.00 | 0.40-1.20 | 0.20-0.80 | 0.10 max. | 0.25 max. | 0.15 | 0.05 | 0.15 | Rem. |
| 6061 | 0.40-0.80 | 0.70 | 0.15-0.40 | 0.15 max. | 0.80-1.20 | 0.04-0.35 | 0.25 max. | 0.15 | 0.05 | 0.15 | Rem. |
| 6070 | 1.00-1.70 | 0.50 | 0.15-0.40 | 0.40-1.00 | 0.50-1.20 | 0.10 max. | 0.25 max. | 0.15 | 0.05 | 0.15 | Rem. |
| 7075 | 0.40 max. | 0.50 | 1.20-2.00 | 0.30 max. | 2.10-2.90 | 0.18-0.28 | 5.10-6.10 | 0.20 | 0.05 | 0.15 | Rem. |

1/ Iron plus silicon, 0.95 maximum.



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TABLE II. Tensile and hardness requirements.

| Alloy and temper | Size, inch | Tensile strength[1], ksi | Yield strength[2], min, ksi | Elongation[3], min., percent | Brinell hardness[4], min. |
|---|---|---|---|---|---|
| 1100-F[5] | - | - | - | - | - |
| 2014-F[5] | - | - | - | - | - |
| -O | All | 30 max. | - | - | - |
| -T4 | All | 55 | 32 | 10 | 100 |
| -T6 | All | 65 | 55 | 6 | 125 |
| 6061-F[5] | - | - | - | - | - |
| -O | All | 22 max. | - | - | - |
| -T4 | All | 26 | 16 | 12 | 50 |
| -T6 | All | 38 | 35 | 6 | 80 |
| -T84 | 0.040-0.093 wall | 35 | 30 | 4 | 75 |
| | 0.094 wall and over | 38 | 35 | 4 | 75 |
| 6070-F[5] | All | - | - | - | - |
| -T6 | All | 48 | 43 | 4 | 95 |
| 7075-F[5] | - | - | - | - | - |
| -O | All | 40 max. | - | - | - |
| -T6 | All | 75 | 65 | 5 | 135 |
| -T73 | All | 66 | 56 | 5 | 125 |

1 Minimum unless maximum is indicated
2 At 0.2 percent offset
3 In 2 inches or 4 times the diameter. Elongation requirements do not apply to material thinner than 0.062 inch
4 500 kg load, 10mm ball. See 3.3.4 for applicability of hardness requirements
5 When mechanical property testing is required or mechanical property limits are negotiated, the temper designation H112 should be used rather than F.





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## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Lot. Unless otherwise specified in the contract or order (see 6.1), a lot shall consist of impacts of the same alloy and temper, the same shape and size, or groups of impacts of similar shape and size as designated by the procuring activity, that have been heat treated in the same furnace charge. If impacts are heat treated in a continuous furnace, impacts charged consequently during continuous operation of the furnace shall be considered a furnace charge.

4.3 Sampling.

4.3.1 For chemical analysis-ingot analysis. At least one sample shall be taken from the molten metal representing one group of ingots poured as a unit from the same source of molten metal. Complete ingot analysis records shall be available to the Government at the ingot producer's facility.

4.3.2 For chemical analysis-finished product analysis. When sampling has not been made in accordance with 4.3.1, one sample shall be taken for each 4,000 pounds or less in each lot in accordance with ASTM E34.

4.3.3 For mechanical property tests. Unless otherwise specified in the contract or order (see 6.1), one tension test specimen or one sample for hardness determination (see 3.3.4) shall be taken from each lot.

4.4 Examination. Each impact shall be carefully examined to determine conformance to this specification with respect to workmanship. Sufficient spot checks shall be made to insure conformance to the tolerances specified. On approval of the procuring activity, a system of statistical quality control may be used for workmanship and dimensional examination.



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4.4.1 Preparation for shipment. Examination of the packaging, packing and marking for shipment shall be made for conformance to the requirements of section 5.

4.5 Test specimens.

4.5.1 Tension tests. Specimens for tensile testing shall conform to the requirements of ASTM B557, and shall be taken from the sidewall of the impact. When practicable, the tension test specimen shall be taken so that its long axis is parallel to the direction of metal flow during extrusion.

4.5.2 Hardness tests. The base of the impact shall be used for hardness testing.

4.6 Test procedures.

4.6.1 Chemical analysis. Chemical analysis shall be made by ASTM E34 to determine conformance with 3.2. A single analysis of a composite sample may be made. In case of dispute, the analysis by ASTM E34 shall be the basis for acceptance.

4.6.2 Tension tests. All tension tests shall be conducted in accordance with ASTM B557.

4.6.2.1 Yield strength. The yield strength shall be determined by the offset method in accordance with ASTM B557.

4.6.3 Hardness test. All hardness tests shall be conducted in accordance with ASTM E10.

4.7 Heat treatment. Specimens in the solution heat treated temper (T4) shall not be required to be tested within 4 days after completion of the heat treatment. If the manufacturer elects, samples may be tested prior to 4 days aging. If, however, the results fail to conform with the requirements of table II, these tests shall be discarded and tests shall be made after the expiration of 4 days without prejudice.

4.8 Rejection and retest.

4.8.1 Rejection. Unless otherwise specified in the contract or order (see 6.1), failure of the samples to meet the requirements of this specification shall be cause for rejection of the lot (see 4.8.2).

4.8.2 Retest. Unless specific retest procedure is specified in the contract or order (see 6.1), two retest samples shall be submitted for each failed sample. Failure of either of the retest samples shall be cause for rejection of the material. Retests shall not be permitted until the supplier has made the necessary corrections in the processing of the material to the satisfaction of the procuring activity.

5. PACKAGING

5.1 Preservation, packaging, and packing. Unless otherwise specified in the contract or order, all impacts shall be preserved, packaged, and packed in accordance with MIL-STD-649.



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5.1.1 When the levels of packaging and packing are not specified, the material shall be packaged and packed in accordance with level C requirements of MIL-STD-649.

5.2 Marking for shipment. In addition to any special marking required in the contract or order (see 6.1), marking for shipment shall be in accordance with MIL-STD-649.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number and date of this specification.
(b) Alloy and temper (see 1.2.1).
(c) Dimensions (see 3.4).
(d) Name of inspection agency when inspection shall be performed by other than the contractor (see 4.1).
(e) If lot is different than 4.2.
(f) If the number of test specimens is different than 4.3.3.
(g) If rejection and retest differ from 4.8.
(h) Level of shipment required (see section 5).

6.2. *Subject term (key word) listing*

Aluminum Alloys
1100-F
2014
6061
6070
7075

Custodian:
Army - MR

Review activities:
Army - AR, AL, EA, ME, MI
DLA - IS

User activities:
Army - ER, CR

Preparing activity:
Army - MR

Project 95GP-A021

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